

DATA301 Food Project

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2026-08-31

Background and Data

Data sets and Sources

Our project focuses on the 2019 release of the Food Access Research Atlas (FARA) (Rhone, 2026) as our main dataset source from the United States Department of Agriculture (USDA)'s Economic Research Service portal. The ERS is a department of the USDA providing economic and agricultural information and statistics.

FARA is a census-tract-level dataset covering the entire United States. As of July 2026, FARA provides two mapping applications: the SNAP-Authorized Retailer Access Map (SRAM) and LRAM (Rhone, 2018). FARA describes food access along these two datasets, considering individual and neighbourhood level resources impacting access, as well as geographic proximity to these stores. During our EDA we have focused only on the 2019 LRAM over using both datasets.

Our data was also supplemented by two additional sources. The US Census Bureau American Community Survey (ACS) 2019 5-Year Estimates for total population and housing units was an additional source of data from the United States Census Bureau (2023). These figures provide tract-level demographic data that more accurately reflects LRAM's 2019 reference period than LRAM's own 2010 Census figures. We also incorporated the USDA Food Environment Atlas (FEA) (U.S Department of Agriculture, 2025), which provided county-level grocery store and fast-food restaurant density estimates (2020 values), extending beyond LRAM's narrower focus on large grocery retailers specifically.

Why This Data set Is of Interest

Food insecurity is a persistent issue facing those around the globe, especially those living in poverty. Food insecurity does not just mean the deprivation of food, but access to healthy and sustainable food as well (Anthem, 2025). Both the USA and New Zealand have faced the consequences of this issue as evidenced both countries having leading rates of obesity, diabetes and heart disease. Notably, heart disease is one of the leading causes of death in both countries (World Health Organization, 2024, 2025).

Some neighbourhoods have supermarkets within easy walking distance, while others are located further from large grocery stores, forcing residents to travel significant distances or rely on smaller, more limited options such as fast food or convenience stores (Rhone, 2018). These "food deserts" are often associated with poverty and reduced diet quality, directly linking to the chronic illnesses noted above (Nulph et al., 2017). LRAM's definition of food access is based solely on distance to supermarkets, supercenters, and large grocery stores and does not capture whether fast food or convenience options are locally available as an alternative. To address this gap, we supplemented our analysis with county-level grocery store and fast-food restaurant density data from the USDA FEA. This data lays the groundwork for future analysis into whether tracts distant from large grocery stores show greater proximity to fast food outlets specifically since fast food consumption is itself commonly associated with minority districts (James et al., 2014) compounding

the relationship between food deserts, systematic inequality and chronic illnesses such as obesity, diabetes, and heart disease.

Our dataset, while based only in the US, helps analyse this issue of equitable distribution of healthy food markets and understand associated trends and patterns. FARA was developed by the USDA to systematically measure this problem on a fine-grained geographic level via census tract - a small subdivision of a country containing about 1,200 to 8,000 residents (Bureau, n.d.). By combining information on population distribution, household income, vehicle access, and distance to the nearest grocery store, LRAM considers both individual and neighbourhood level resources impacting access, as well as geographic proximity to food retailers. This allows us to identify which areas are more affected by low food access and explore what characteristics such as income, geography, or demographic composition are associated with this disadvantage.

This dataset can help identify which regions of the United States have the highest concentrations of low-access communities, which populations are disproportionately affected, and what factors may contribute to this disparity. As explored later in this report, these disparities appear to reflect a multi-faceted interaction between distinct factors including geography, income, and racial/ethnic composition rather than a direct correlation with any single factor.

Data Types and Structure

Our merged dataset is organised as a table, where each row represents one U.S. census tract; the primary unit we are analysing. County-level FEA variables are also included, joined via each tract's county identifier, meaning tracts within the same county share identical values for these specific variables. It contains 72,531 census tracts along 50 variables. The dataset has both numerical, categorical, and logical variables. Most of our dataset are numerical variables such as population counts, poverty rate, median family income, low-access percentage, vehicle access percentage, and food store measures. Our primary outcome variable throughout this analysis, `low_access_pct_1_10`, represents the percentage of a census tract's population living more than 1 mile (in urban areas) or 10 miles (in rural areas) from the nearest supermarket, supercenter, or large grocery store, calculated as `LAPOP1_10` divided by 2010 tract population, multiplied by 100. There are a smaller number of categorical variables, such as whether a tract is urban or rural, income tier groups, race/ethnicity percentage bins. Logical/flag variables are also included to indicate completeness and merge success across the integrated datasets.

The data is not a time series dataset, because it does not follow each census tract over many years. However, the merged dataset does include variables from different years. For example, the population count in the LRAM is based on 2010 values, the ACS population and housing variables are from 2019, and the FEA store variables are from 2020. We treated these as contextual variables drawn from different sources rather than as repeated yearly observations of the same variable.

The dataset does not include latitude and longitude coordinates. Instead, location is shown through census tract IDs, county FIPS codes, state, and county names. It also does not have diagnostic categories like a medical dataset would. However, it does include classification variables, such as whether a tract is urban or rural, low-access or not, low-income or not, and whether it has low vehicle access.

Completeness

The final merged dataset has 72,531 census tracts. Most of the main Food Access Research Atlas variables were complete. Notably, our outcome variable, `low_access_pct_1_10`, had no missing data. Notable variables that had missing values were `poverty_rate` was missing for 3 tracts, `median_family_income` was missing for 748 tracts, and the race/ethnicity percentage variables such as `pct_black` and `pct_hispanic` were missing for 4 tracts. The vehicle access variable also had 162 invalid or missing values after we calculated `pct_no_vehicle`. Given the low proportion of missing values across most variables (under 0.3%), and the lack of a reliable method to impute suppressed Census estimates, we used complete-case analysis throughout. We excluded tracts with missing values only from the specific comparisons involving those variables, rather than imputing or removing them from the dataset entirely.

The FEA variables had more missing values. After we changed special missing-value codes like -9999 and -8888 into NA, `grocery_stores_per_1000_2020_county` was missing for 2,689 tracts and `fast_food_restaurants_per_1000_2020_county` was missing for 875 tracts. These missing values were from the FEA variables data source themselves, not from the urban/rural variable. The missing grocery-store data appeared in both rural and urban tracts, but was disproportionately concentrated in rural tracts (2,145 of 2,689 missing tracts – 79.8%)

Dataset Integration

Since we used more than one dataset, we had to clean the IDs before merging, making sure census IDs matched. We converted the LRAM `CensusTract` variable into an 11-character string, so leading zeros wouldn't be lost. This was important because if census tract IDs are read as numbers, the zeros at the start could disappear resulting in a failing merge. After fixing this, we merged the ACS 2019 population and housing data using the census tract ID.

For the FEA data, the merge was done at the county level instead of the tract level. We made a `county_fips` variable using the first five digits of the census tract ID, then used that to merge in the grocery-store and fast-food restaurant variables. This means all tracts in the same county have the same FEA values, so we treated those variables more as background context rather than as the main tract-level predictors.

After merging, we checked whether the joins worked. There were no unmatched rows for the ACS population data, ACS housing data, or FEA county merge. We did not delete rows from the cleaned dataset. Instead, missing or invalid values were kept as NA, and we added flags such as `missing_core_analysis_flag`, `missing_merged_variables_flag`, and `invalid_vehicle_pct_flag` so we could filter them later if needed.

Ethics, Privacy and Security

Ethical considerations

This project did not require formal ethics approval. All three datasets are secondary data and are already available to the public. FARA, the ACS and the FEA are all published by USA Government Agencies with none of the data being collected from people directly. The closest we get to any individual is the census tract, which contains population groups between ~1,200 to ~8,000 individuals (Bureau, n.d.), keeping them safe and protected from identification within the datasets.

Within the data itself FARA and FEA don't contain raw census counts. Instead, they are USDA constructed classifications with decisions layered on top. This shows most in the distance thresholds used to define 'low access' (1 mile in Urban Tracts which shifts to 10 miles in Rural Tracts (Economic Research Service, U.S. Department of Agriculture, n.d.)). This means the State ranking is more an artefact of how rural a state is, not how underserved it is. A more rural State will generally rank higher regardless of actual food access, so we had to be careful not to present that ranking as a clean measure of deprivation.

Figure 6 and table 4 both check low food access with race/ethnicity composition and income tier. This kind of inference has a history of being read as if the demographic composition is itself responsible for the outcome, in turn, the more supported explanation is historical redlining, lack of investment and retail location decisions. (George & Tomer, 2021; Zenk et al., 2005). Any race/income association is reported as correlated and we note the structural explanation, rather than letting the tables speak for themselves.

Finally, as a group based in New Zealand analysing a USA dataset part of why we selected the FARA dataset was that food insecurity and its associated outcomes matter in a NZ context too. This doesn't mean the patterns we find here would translate to New Zealand though given our differences in population, ethnic diversity, policy and settlement patterns. We're treating our findings as illustrations of the US case rather than using it to directly inform decisions based around New Zealand's food access.

Privacy Concerns

None of the datasets used contain individual level records. FARA, ACS and FEA are all published at a broader level, that being census tract and county level. The Census Bureau applies disclosure avoidance methods; suppression, modification and swapping to prevent identification of any individual from ACS data. (U.S. Census Bureau, n.d.). The risk of identifying any specific individuals from this data is very low by design before we have even touched it.

Two observations are still prudent, firstly, cross-tabulating variables from the three sources together as we did in Figure 6 and Table 4, (race/ethnicity bin x income tier x urban/rural x state) produces far more in-depth and specific outputs than any single source table on its own. This does not push below the geographic resolution already present in each source, meaning we are not adding to any identification risk.

Finally, this wasn't something we restricted to the EDA stage, Figure 5 and Table 4 name specific states next to low-access and demographic statistics because that made the clearest plot, not because we weighed how identifiable or characterised that made those places. No individuals are named, but real, named places are being associated with poverty, low access and race statistics, which is noteworthy even though it isn't technically a privacy breach.

Security of Project Data and Results

Our analysis, code, cleaned and merged datasets, and report drafts are all stored in a private GitHub repository which is accessible only to the three of us. This mattered mainly for the derived data that we created as the raw datasets we used are all publicly available as it stands but our merged and cleaned datasets as well as any resulting graphs or inferences are not. By using a private repository, we kept from distributing any product beyond what the agencies themselves released.

With more time, resources and hindsight, there are a few practical steps we could add. We would turn on GitHub access logging so there's a record of who changed what and when, since currently any of the three of us can edit the repository with no audit trail. We would also write a short data file documenting where each raw source was obtained, on which date and what transformations were applied to it. This is about manageability and reproducibility as well as security, but it directly supports the project. Finally, after the submission for both our group and individual reports are done, we would set an archival or deletion date for our files to prevent any unwanted access further down the line.

Exploratory Data Analysis

Variable Types

Table 1: Table 1: Types of Key Variables Used in Analysis

Variable	Type
low_access_pct_1_10	Numeric
poverty_rate	Numeric
pct_no_vehicle	Numeric
pct_black	Numeric
pct_hispanic	Numeric
urban_rural	Categorical
income_tier	Categorical
grocery_stores_per_1000_2020_county	Numeric
fast_food_restaurants_per_1000_2020_county	Numeric
acs_population_match_flag	Logical/Flag
food_env_county_match_flag	Logical/Flag

Summary Statistics

Table 2: Table 2: Summary Statistics for Key Variables

Variable	Mean	Median	SD	Missing
Low Access (%)	21.27	1.68	30.95	0
Poverty Rate (%)	15.18	11.95	11.92	3
No Vehicle (%)	9.4	5.29	12.34	162
% Black	13.84	3.99	22.3	4
% Hispanic	15.27	6.21	20.83	4
Grocery Stores/1000	0.2	0.17	0.12	2689
Fast Food/1000	0.77	0.78	0.18	875

Table 3: Table 2b: Distribution of Tracts by Urban/Rural Status

urban_rural	Count	Percent
Rural	17362	23.9
Urban	55169	76.1

The merged data set contains 72,531 census tracts across 50 variables.

Table 1: Summarises the type of each key variable used in analysis such as the numeric variables including the outcome (% of tract population with low food access), economic and vehicle access controls, race/ethnic composition and county level retail density. Categorical variables include urban/rural status and income tier. Logical/flag variables indicate whether each tract successfully matched across merged data sources.

Table 2: Shows summary statistics for numeric variables. Low food access, % Black, and % Hispanic show notably higher standard deviations compared to their medians, consistent from the right skewed data examined in the Spearman correlation matrix (Figure 1). Missing data was small for most variables (>0.3%), while the two Food Environment Atlas variables show higher missing values (3.7% for Grocery Stores, and 1.2% for Fast Food), as explained in the “Completeness” section above.

Table 3: Distribution of tracts by Urban/Rural classification. There are 17,362 rural tracts (34.9%) and 55,169 Urban tracts (76.1%), providing context for Urban/Rural comparison in Figure 4.

Correlation of Key Variables

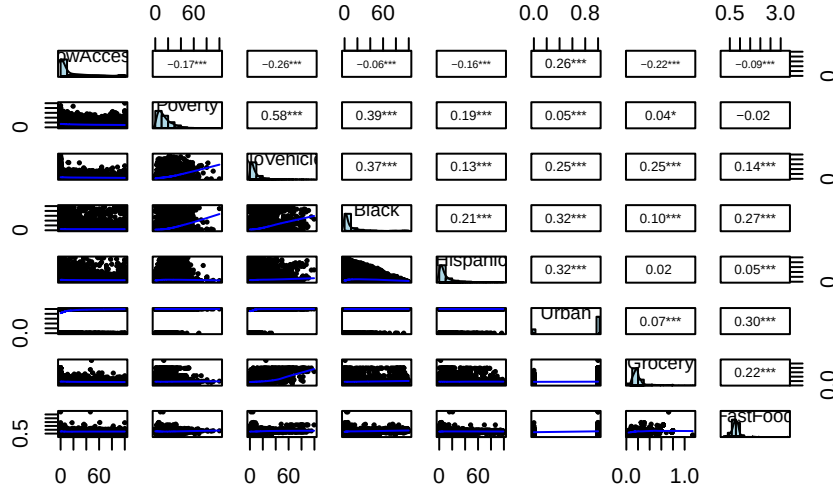


Figure 1: Spearman correlation matrix of key variables: food access outcome, economic and vehicle access controls, racial/ethnic composition, and food retail context.

Figure 1 shows a pairwise Spearman correlation matrix across seven key variables. This was computed on a sample of 5,000 census tracts (from the full 72,531 tracts) to improve rendering performance. The variables shown are: LowAccess (% of tract population beyond the low-access distance threshold), Poverty (tract poverty rate), NoVehicle (% of households without a vehicle), Black and Hispanic (racial/ethnic composition), Urban (binary urban/rural indicator), Grocery, and FastFood (county-level retail density per 1,000 residents).

Spearman correlation was used rather than Pearson due to the high amount of right-skewed distributions observed in several variables. This shows vehicle access has the strongest association with low food access ($p = -0.26$) among all variables considered, followed by poverty rate ($p = -0.16$). Racial/ethnic composition shows weaker individual associations ($p = -0.05$ for percentage Black, $p = -0.15$ for percentage Hispanic), suggesting that no single variable strongly predicts low food access on its own, a pattern investigated further in Figure 6 and Table 4’s comparison. Both Food Environment Atlas variables (grocery store and fast-food density) show modest associations with low food access ($p = -0.20$ and $p = -0.18$ respectively), providing initial support for their inclusion as supplementary context.

Distribution of Low Food Access

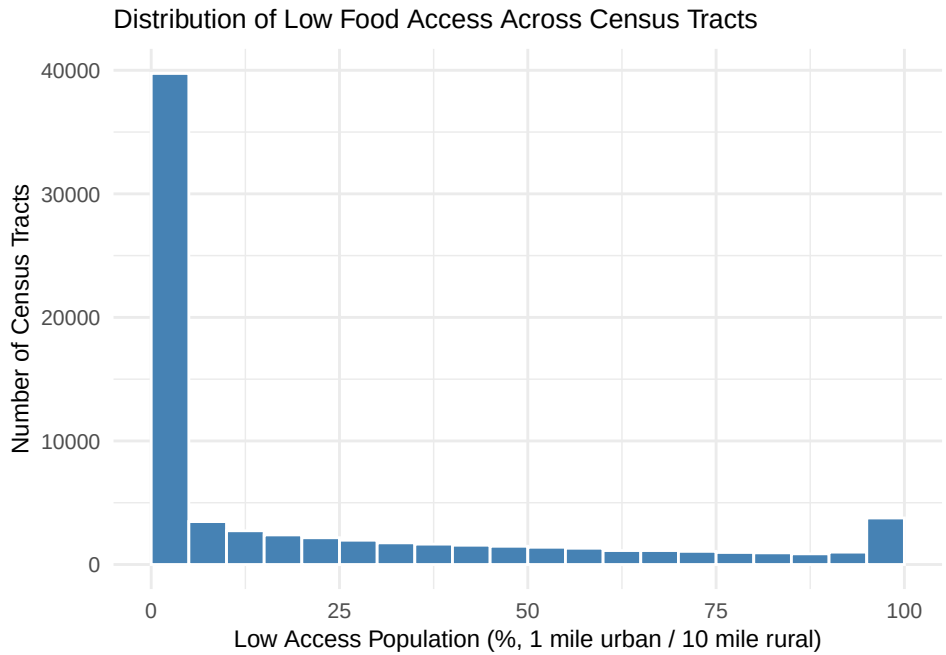


Figure 2: Distribution of Low Food Access Across Census Tracts

Poverty Rate vs Low Food Access

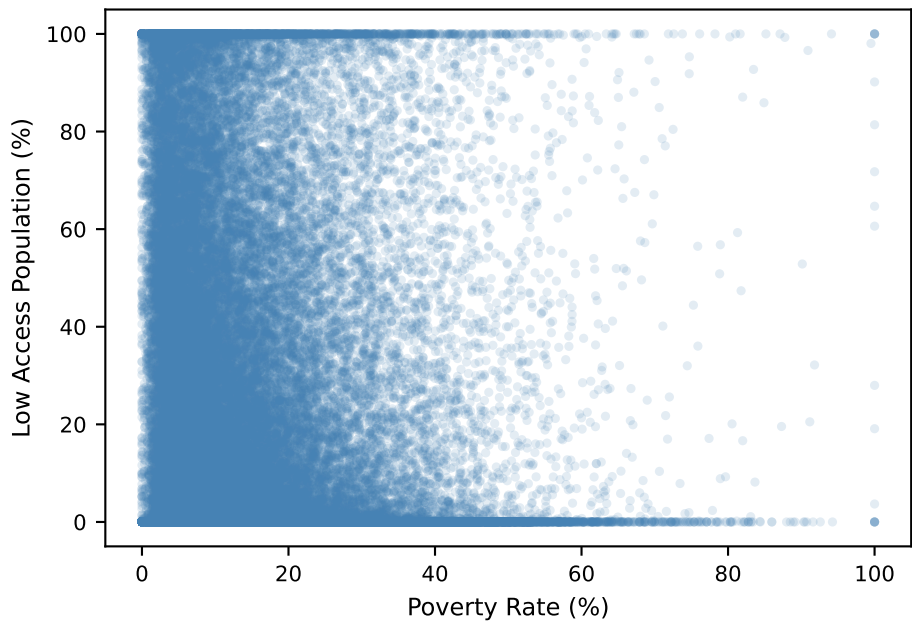


Figure 3: Poverty Rate vs Low Food Access by Census Tract

Figure 2 shows the distribution of Low Access Population (% , 1 mile urban / 10 mile rural) across all 72,531 census tracts. The distribution is heavily right skewed, the graph shows over half of the tracts show none, or very minimal amounts of those considered a low-access population, while a smaller number show

high percentages, including another cluster near 100%. This indicates that low food access is concentrated in a small subset of tracts rather than being a widespread issue across the US. This has motivated our further investigation into what distinguishes this subset presented in the remainder of the EDA.

Figure 3 shows the relationship between Poverty Rate (%) (share of tract’s population living below federal poverty threshold) and low food access. Interestingly, despite common assumption being that a primary driver of food access disparities being related to poverty, the figure shows no strong visual pattern. The graph shows that at every poverty level, from 0% to 100%, span the full range of low access outcomes. This matches the correlation matrix (Figure 1), that showed a comparatively weak association between poverty and low food access ($p = -0.16$). This has motivated further investigation into other potential factors shown in following figures.

Low Food Access by Urban/Rural Status

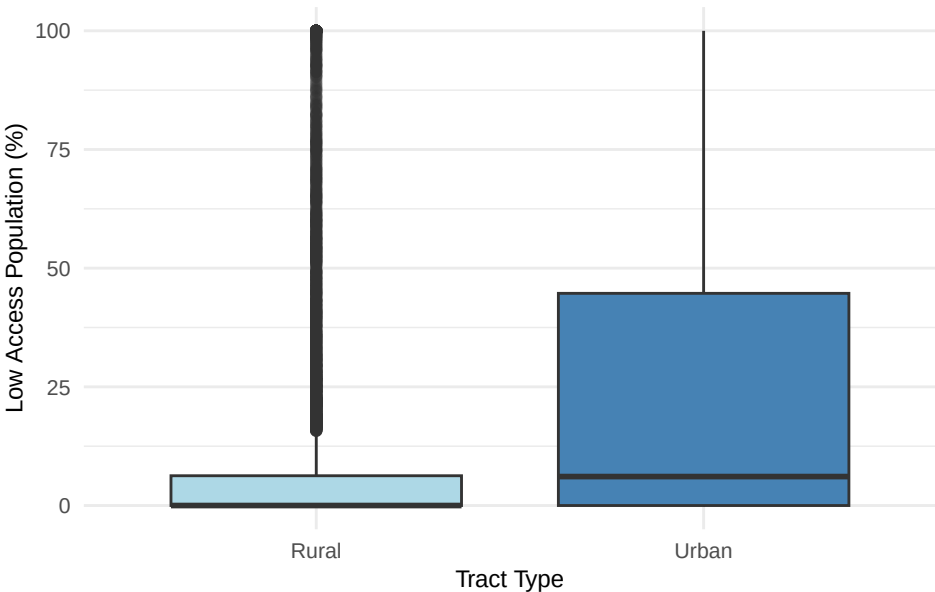


Figure 4: Low Food Access by Urban/Rural Status

Low Food Access by State

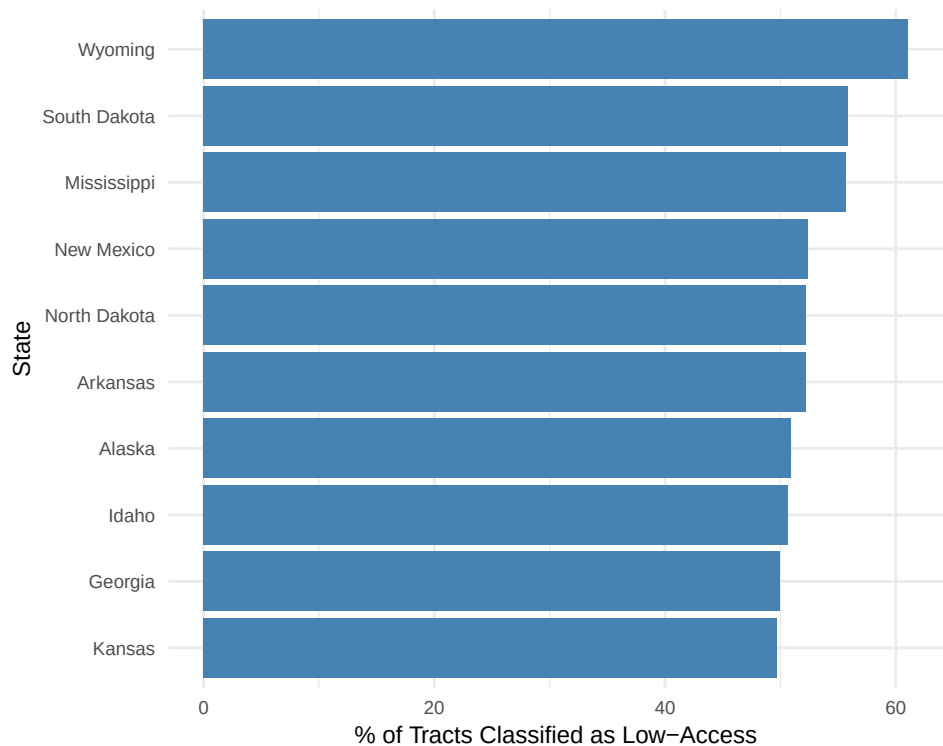


Figure 5: Top 15 States by Share of Tracts Meeting USDA Low-Access Criteria

Figure 4 compares low food access between urban and rural census tracts. The `urban_rural` variable classifies tracts using the Census Bureau’s urbanized area definitions, and directly determines which distance threshold applies to (1 mile for urban tracts, 10 miles for rural tracts). Rural tracts show a lower median low-access percentage (approximately 2-3%) than urban tracts (approximately 7%), but a substantial number of rural tracts are extreme outliers at or near 100% low access. This likely reflects some remote communities will be far from any large grocery store. Urban tracts, by contrast, show a wider interquartile range (about 0-45%), suggesting moderate levels of low access are more common in urban areas, while extreme, near-total low access is rare. This pattern in rural tracts may help explain the secondary cluster near 100% observed in the overall distribution (Figure 2).

Figure 5 shows the ten states with the highest share of census tracts meeting USDA’s official low-access classification criteria (at least 33% of the tract population, or 500 people, living beyond the 1 mile/10 mile distance threshold). Sparsely populated Western and Midwestern states, Wyoming, South Dakota, New Mexico, North Dakota, Alaska, and Idaho, show high low-access shares consistent with geographic remoteness supported by the rural outlier pattern observed in Figure 4. Several Southern states, Mississippi, Arkansas, Georgia, also appear on this list; cross-referencing these states against racial composition (Table 4) reveals notably higher average Black population share (18.5%-41.0%) compared to the national tract-level average (13.8%, Table 2), while the Western/Midwestern states on this list show minimal Black population share (under 3%). This suggests geographic remoteness and a separate pattern potentially connected to the race/income interaction (examined in the following figure) drive state level low food access.

Low Food Access by Race/Ethnicity Composition and Income Tier

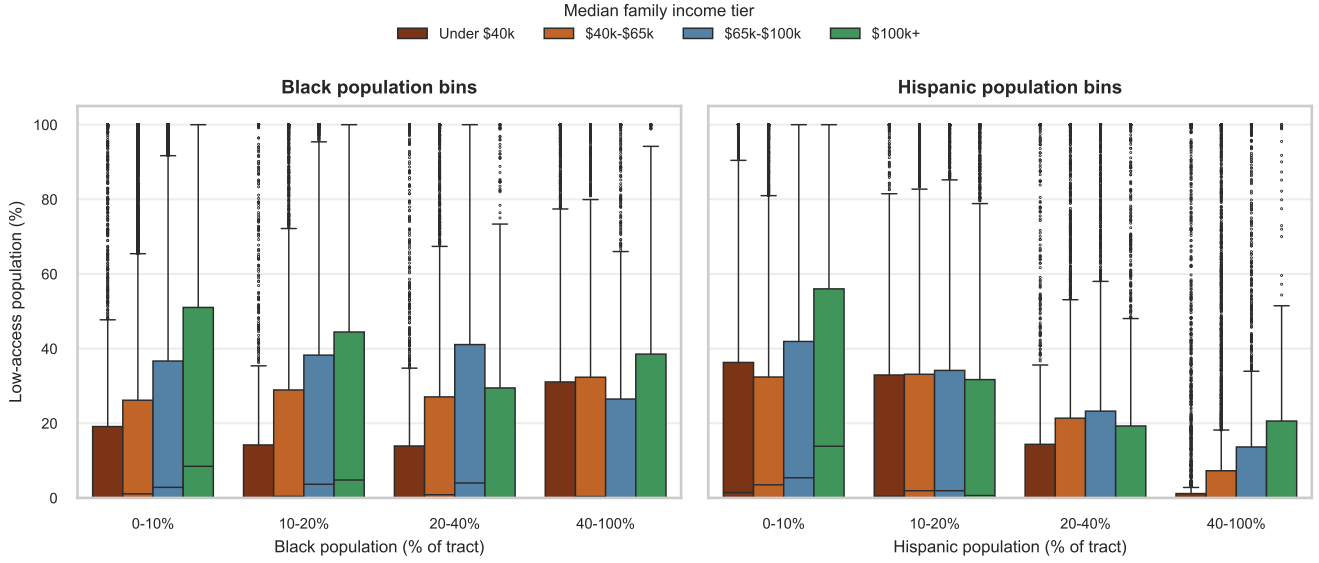


Figure 6: Low Food Access by Race/Ethnicity Composition and Income Tier

Table 4: Racial Composition of High-Low-Access States

state	Black_pct	Hispanic_pct
Mississippi	41	2.8
Georgia	32.1	8.2
Arkansas	18.5	5.8
Kansas	6.9	11
Alaska	2.8	5.1
New Mexico	2	44.5
South Dakota	1.2	2.6
North Dakota	0.9	2
Wyoming	0.9	9
Idaho	0.6	11

Figure 6 shows that low food access by racial/ethnic composition and income tier. `Black_pct_bin` and `hispanic_pct_bin` classify tracts by Black or Hispanic population share (0-10%, 10-20%, 20-40%, 40-100%), split by `income_tier`. This shows whether race is associated with low food access once income is held approximately constant, addressing the weak bivariate relationship (Figures 1 and 3).

This showed differing results. Among Hispanic concentrated tracts, median low access rates decline as concentration increases across all tiers. For Black concentrated tracts, low income tracts (under \$40k) with high population share (40-100%) show an elevated median (around 31%), the highest of any group, differing from the Hispanic tracts. This suggests race and income interact specifically for Black communities consistent with the previous literature on structural drivers of food access inequality (James et al., 2014).

Table 4 shows the racial composition of the 10 states from Figure 5. Mississippi, Georgia, and Arkansas show notably higher Black population share (18.5%-41.0%) than the national average (13.8%, Table 2), while Alaska, the Dakotas, Wyoming, and Idaho show minimal Black share (under 3%) this supports two distinct mechanisms behind these states' rankings: geographic remoteness, and a pattern connected to the race-income interaction in Figure 6. New Mexico is a notable exception, combining high low-access with a high Hispanic share (44.5%), contrasting with the tract-level finding that Hispanic-concentrated tracts generally show lower low-access (Figure 6).

Individual Contributions

References

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